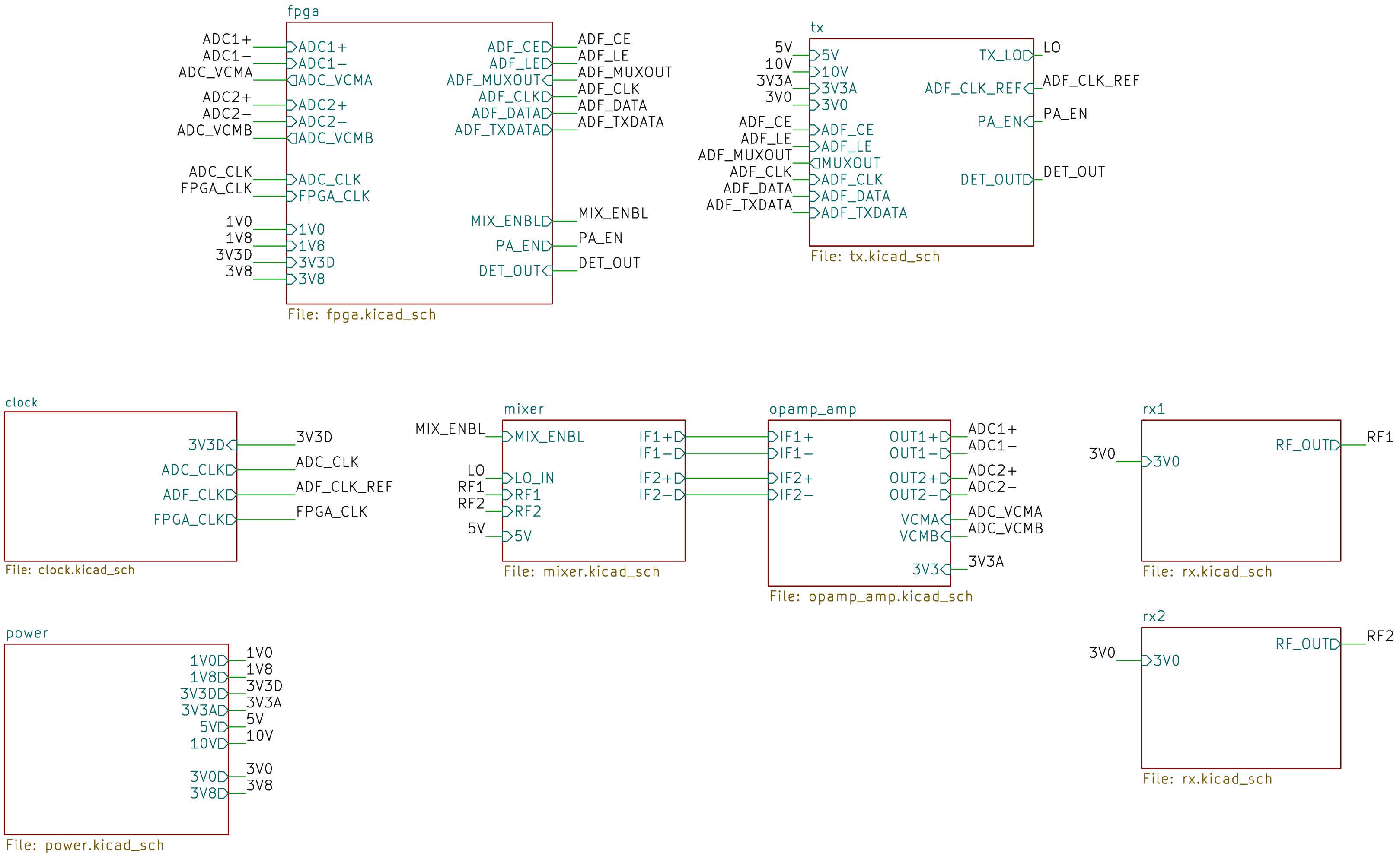
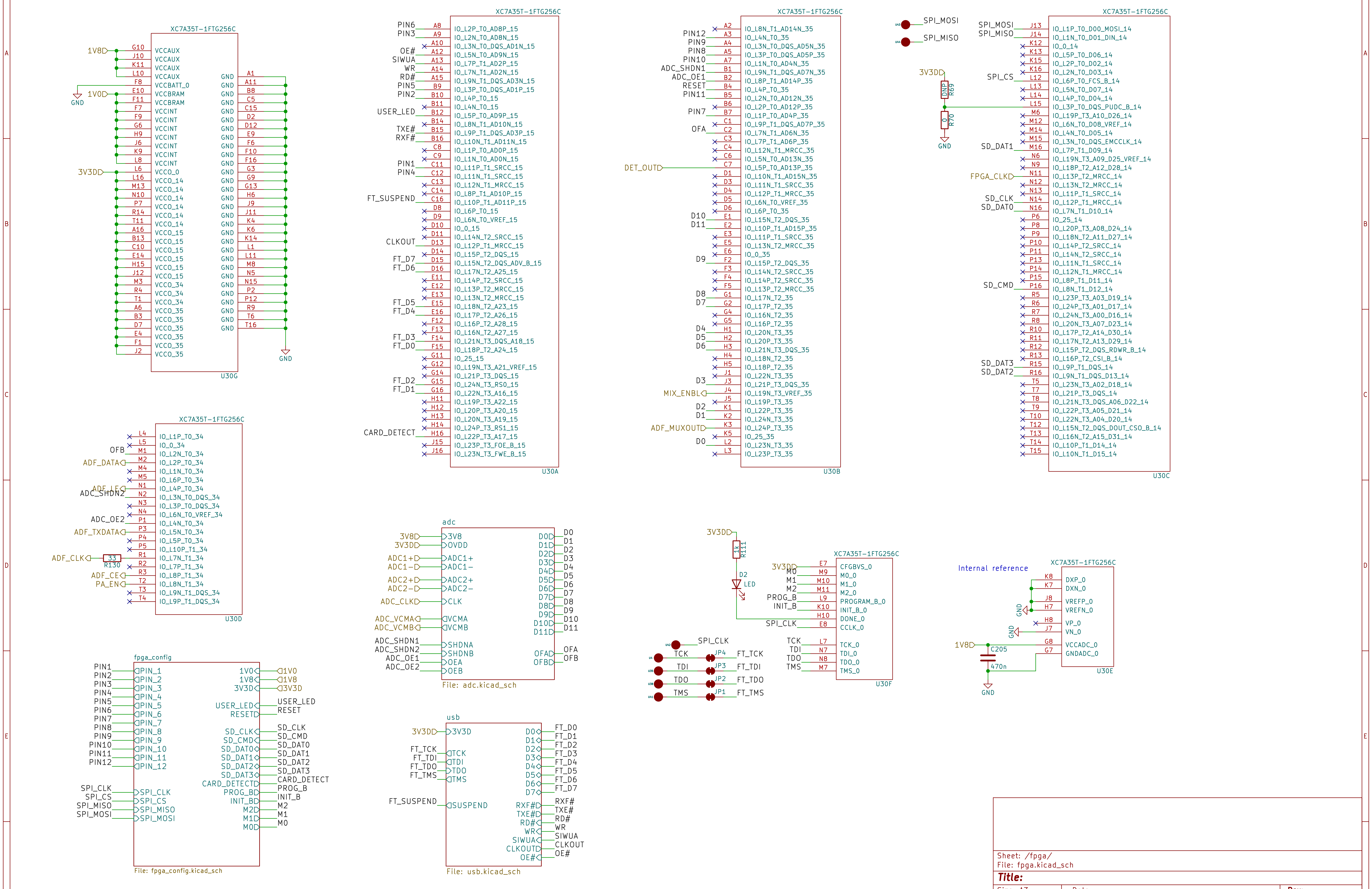


Sheet: /
File: fmcw3.kicad_sch

Title: FMCW3 by CENK KESKİN

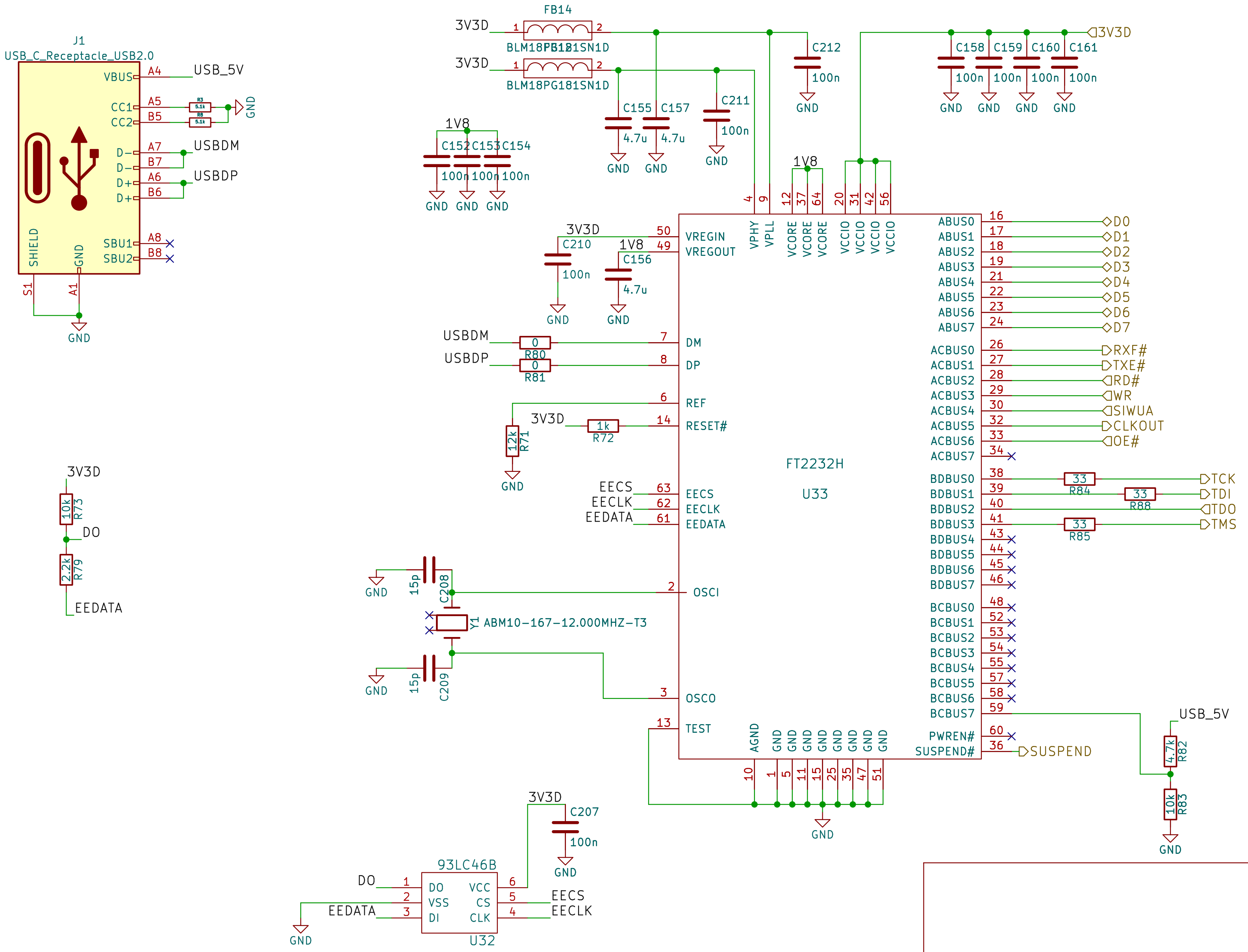
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/12



Sheet: /fpga/
File: fpga.kicad_sch

Title:

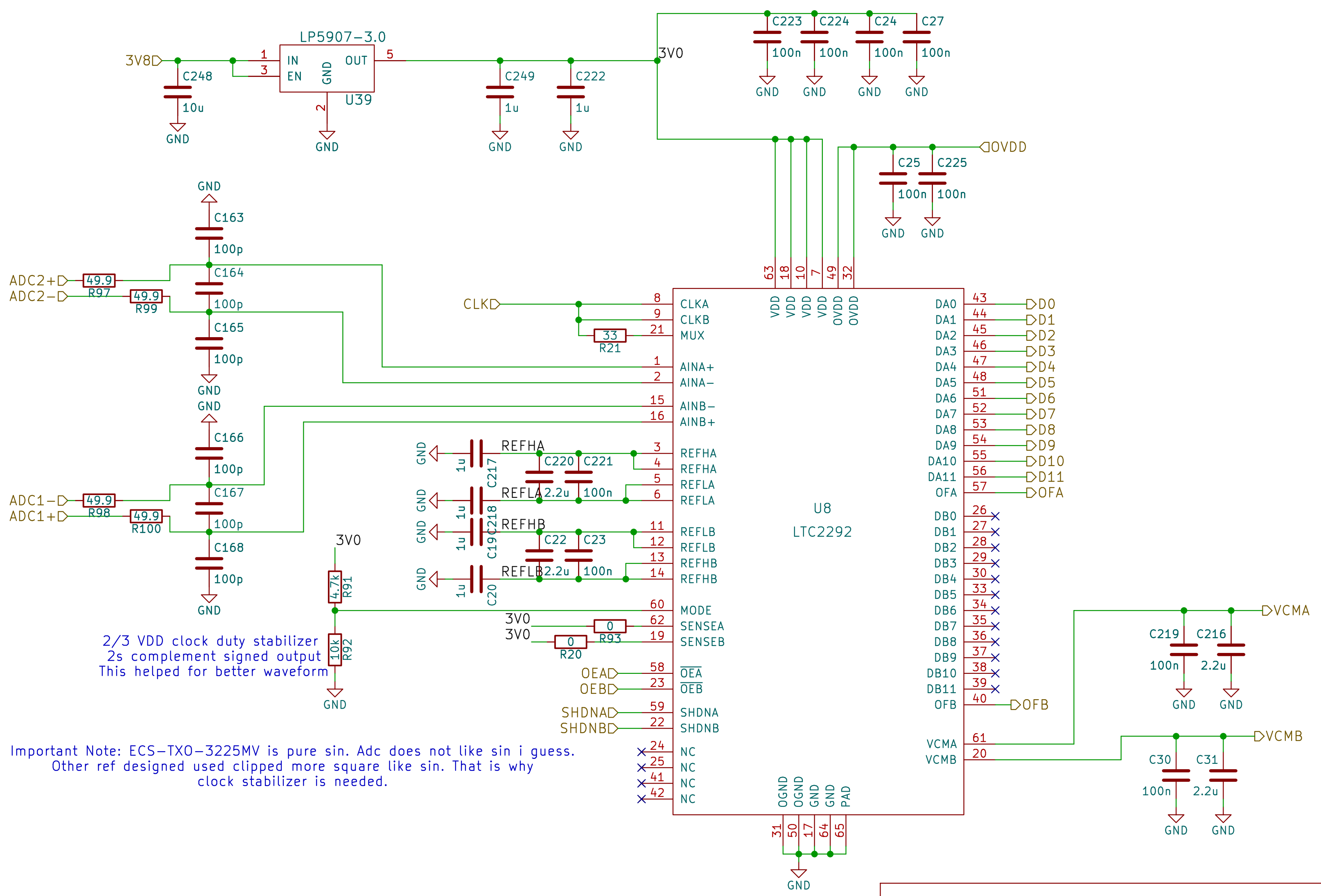
Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 2/12



Sheet: /fpga/usb/
File: usb.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 3/12

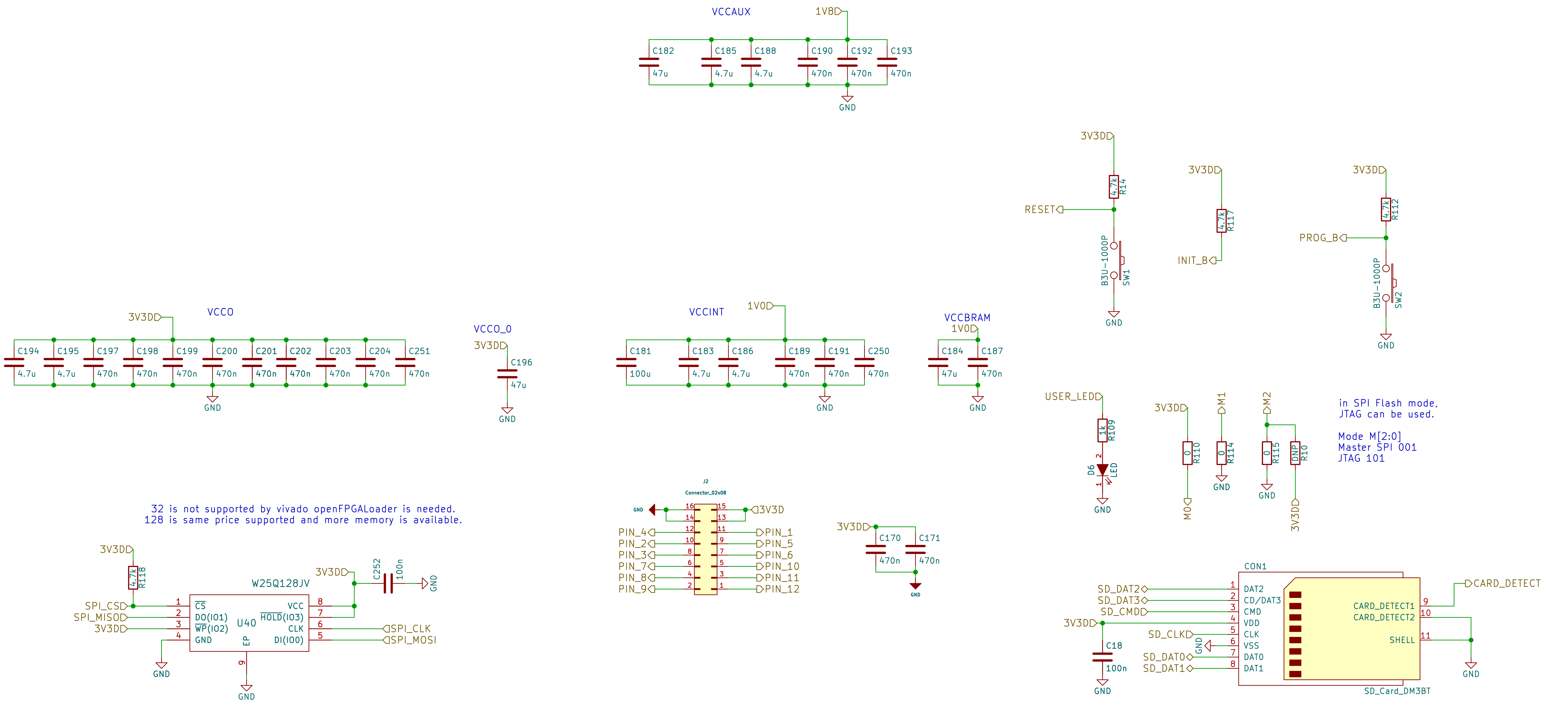


Sheet: /fpga/adc/
File: adc.kicad_sch

Title:

Size: A4 Date: Rev:

KiCad E.D.A. 9.0.7 Id: 4/12

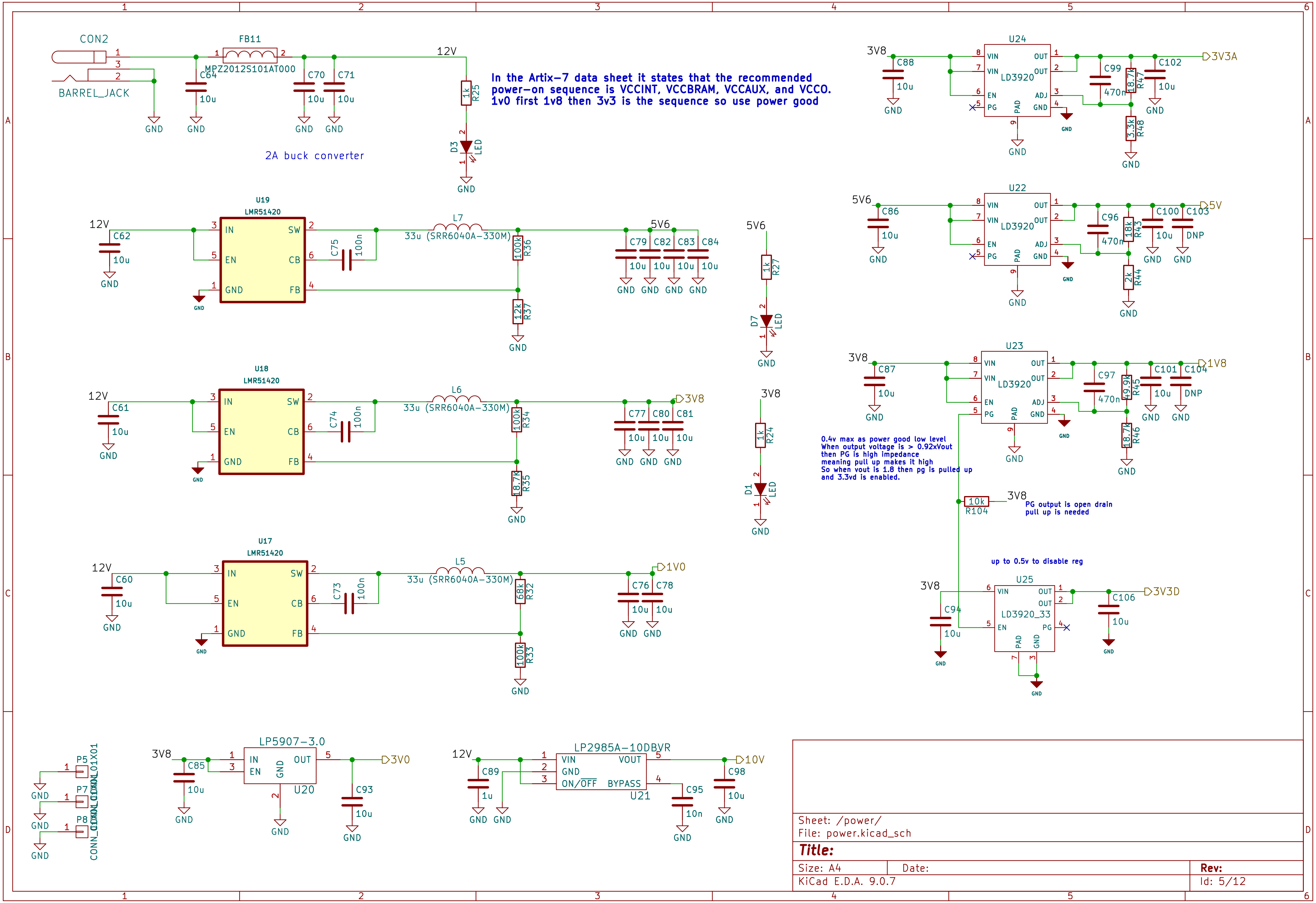


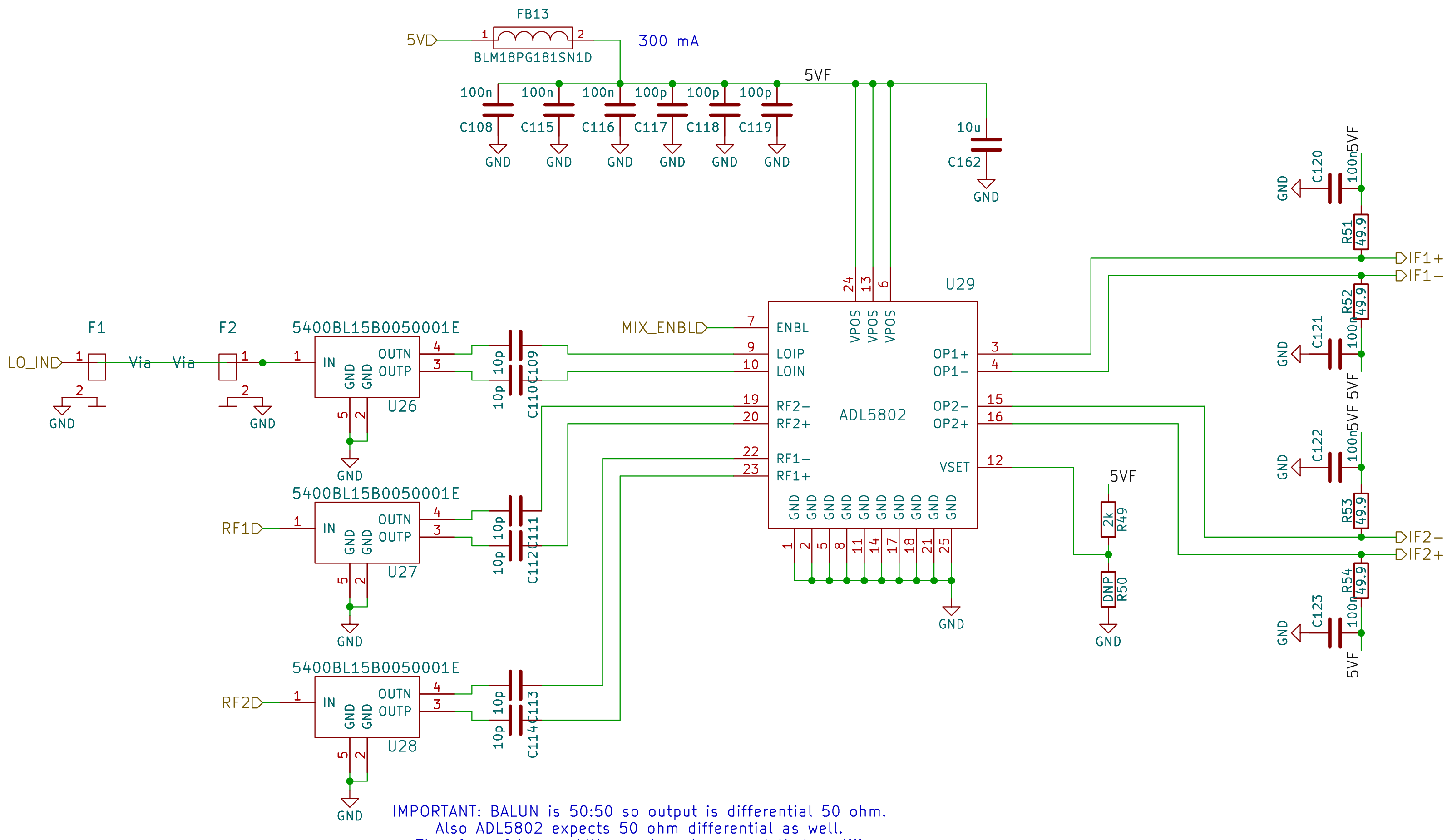
Sheet: /fpga/fpga_config/
File: fpga_config.kicad_sch

Title:

Size: A3 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 12/12


In the Artix-7 data sheet it states that the recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO. 1v0 first 1v8 then 3v3 is the sequence so use power good
2A buck converter
CON2
BARREL_JACK
FB11
MPZ2012S101AT000
12V
C64 10u
C70 10u
C71 10u
R25 1k
D3 LED
U19 LMR51420
L7 33u (SRR6040A-330M)
C75 100n
R36 100k
R37 12k
5V6
C62 10u
C79 C82 C83 C84 10u
U18 LMR51420
L6 33u (SRR6040A-330M)
C74 100n
R34 100k
R35 18.7k
3V8
C61 10u
C77 C80 C81 10u
U17 LMR51420
L5 33u (SRR6040A-330M)
C73 100n
R32 68k
R33 100k
1V0
C60 10u
C76 C78 10u
R27 1k
D7 LED
R24 1k
D1 LED
U24 LD3920
3V3A
C88 10u
C99 470n
R47 18.7k
C102 10u
R48 3.3k
U22 LD3920
5V
C86 10u
C96 470n
R43 18k
C100 10u
C103 DNP
R44 2k
U23 LD3920
1V8
C87 10u
C97 470n
R45 49.9k
C101 10u
C104 DNP
R46 18.7k
0.4v max as power good low level
When output voltage is > 0.92xVout
then PG is high impedance
meaning pull up makes it high
So when vout is 1.8 then pg is pulled up
and 3.3vd is enabled.
R104 10k
PG output is open drain
pull up is needed
up to 0.5v to disable reg
U25 LD3920_33
3V3D
C94 10u
C106 10u
LP5907-3.0
U20
3V0
C85 10u
C93 10u
LP2985A-10DBVR
U21
10V
C89 1u
C95 10n
C98 10u
P5 P7 P8 CONN_01X01
GND
Sheet: /power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 5/12



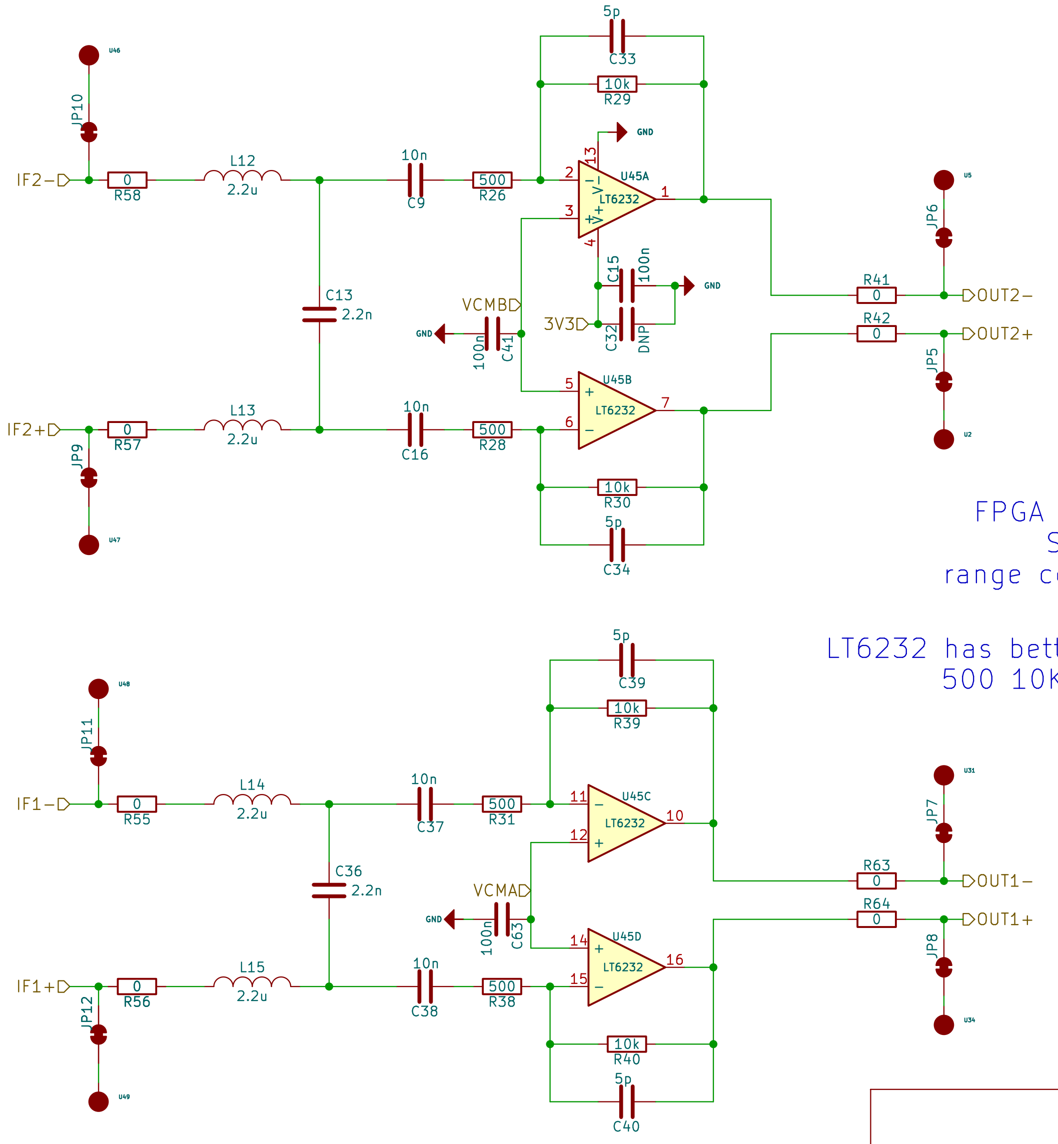
IMPORTANT: BALUN is 50:50 so output is differential 50 ohm.
Also ADL5802 expects 50 ohm differential as well.
Therefore rf trace width spacing changes at that part!!!

Sheet: /mixer/
File: mixer.kicad_sch

Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 9.0.7 Id: 6/12



FPGA FIR Filter works perfectly.
Second stage for HPF
range compensation filer is removed

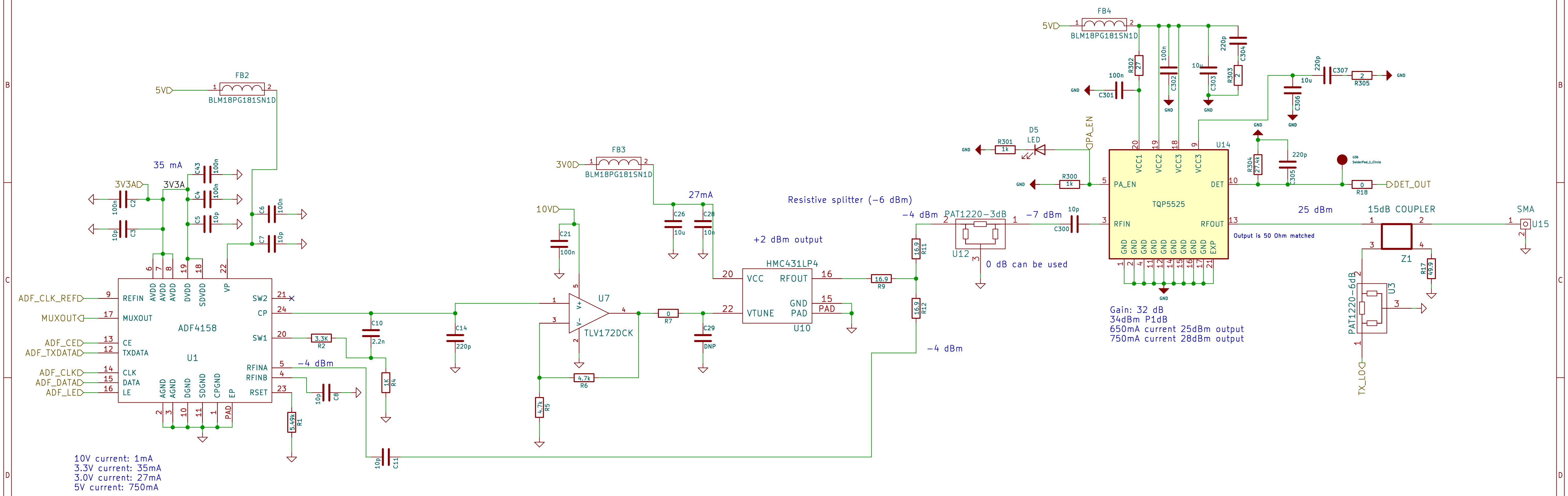
LT6232 has better noise performance as 1.1nV/√Hz
500 10K has less noise contribution

Sheet: /opamp_amp/
File: opamp_amp.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.7 | Id: 7/12

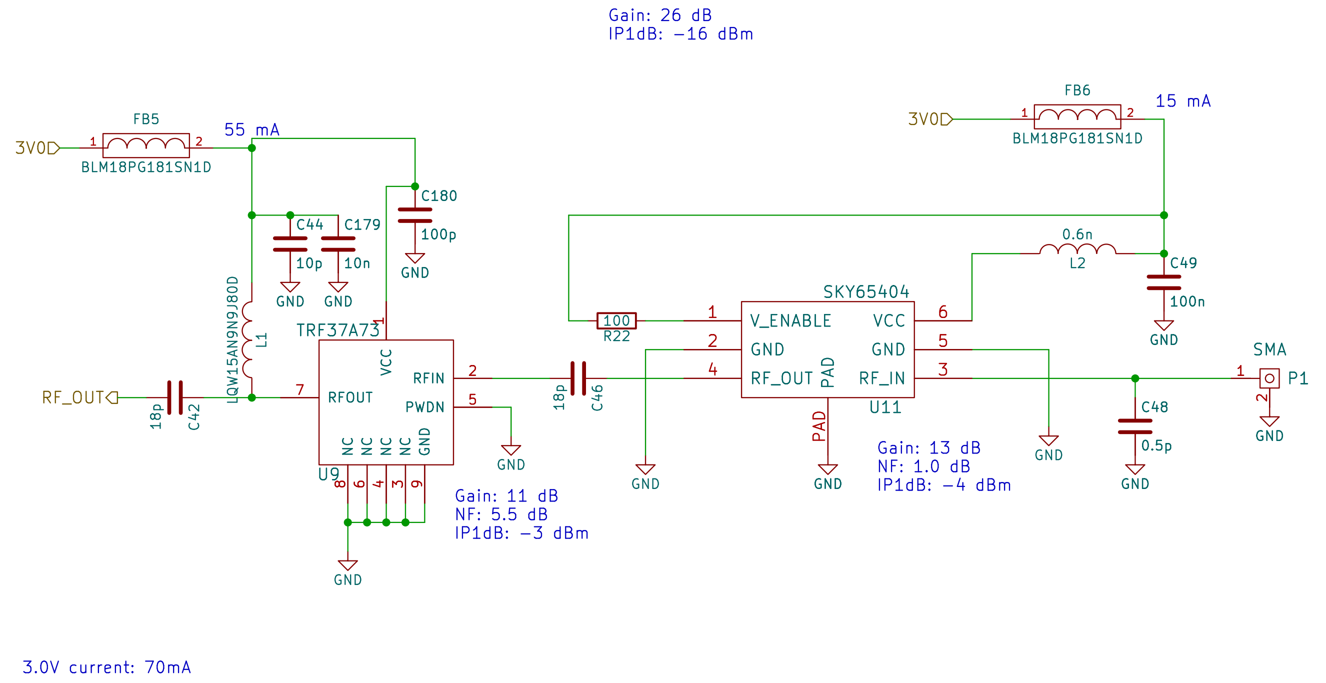


Sheet: /tx/
File: tx.kicad_sch

Title:

Size: A3 Date: Rev:

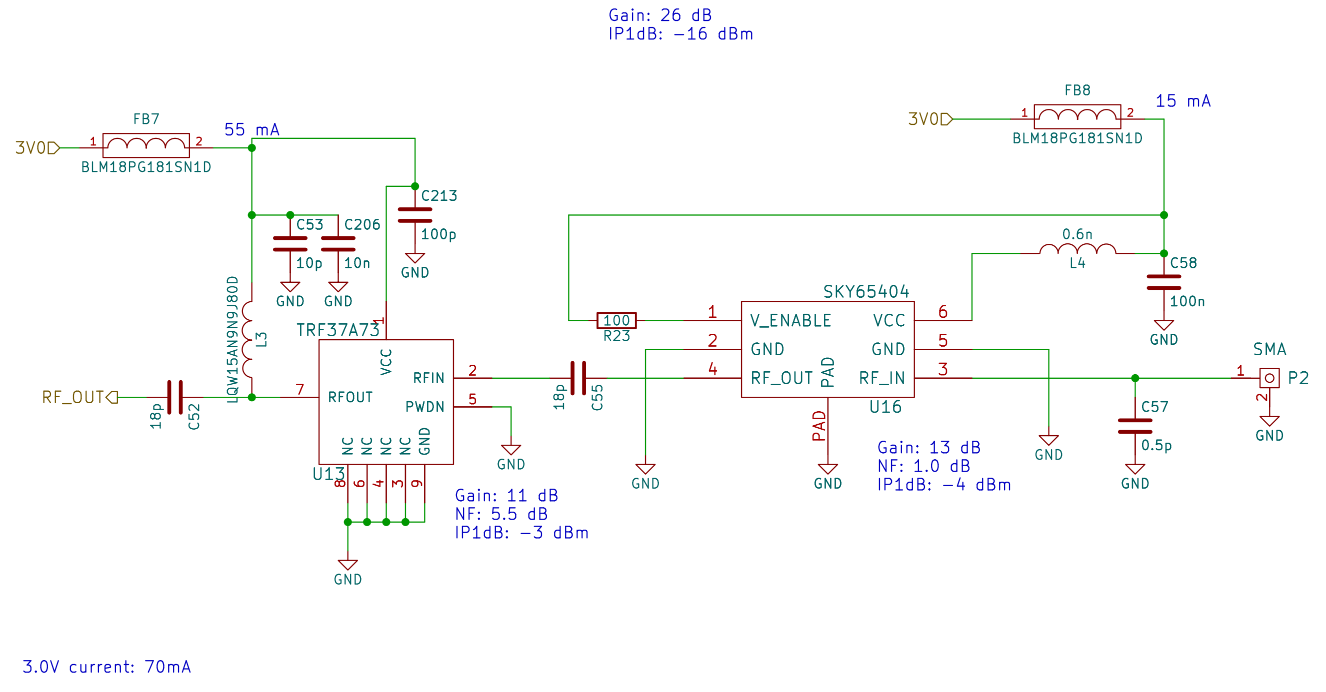
KiCad E.D.A. 9.0.7 Id: 8/12



Sheet: /rx1/
File: rx.kicad_sch

Title:

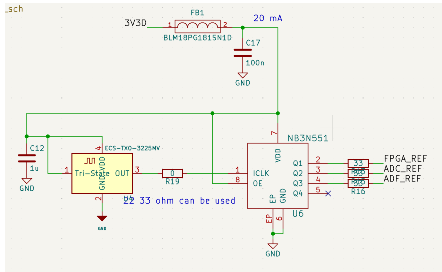
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 9/12



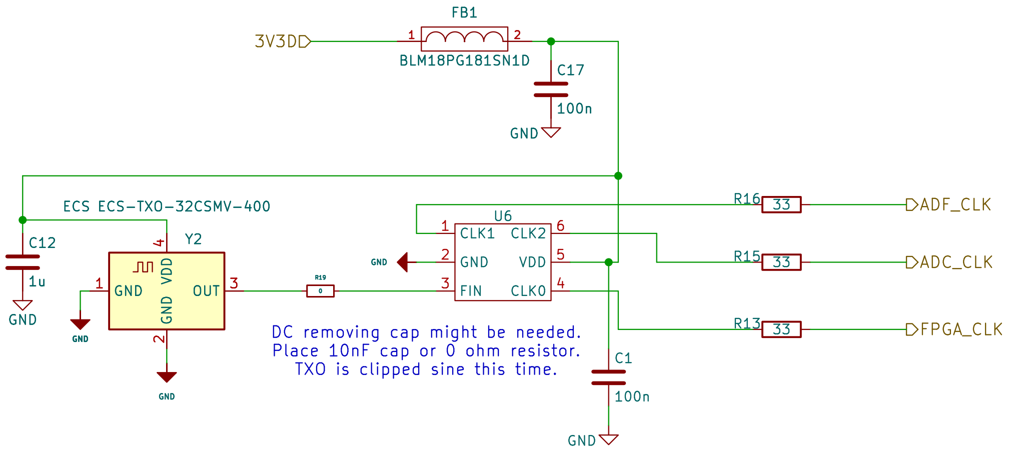
Sheet: /rx2/
File: rx.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 10/12



This is the prev design:
NB3N551 is obs.
Also this txo is generating pure sin.
ADC might need cmos like input not sin.
That could be the reason of duty stabilizer.



Sheet: /clock/
File: clock.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.7 | Id: 11/12